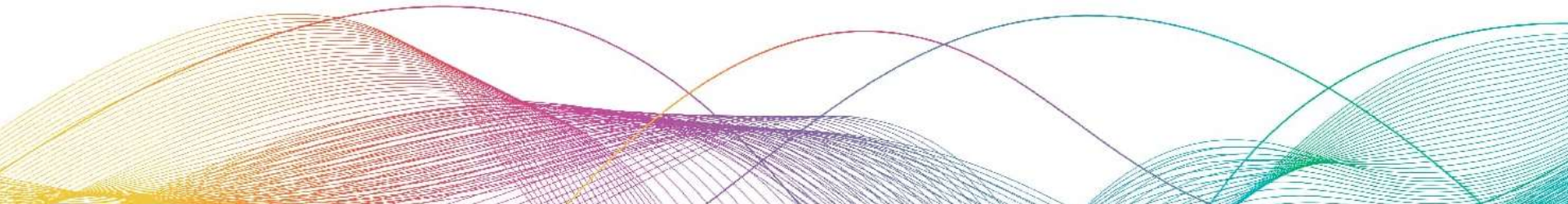


08

Call API To Create A Knowledge System



Solar System

Overview:

This project is a Solar System 3D interactive exploration platform based on Three.js + Flask, supporting features such as 3D celestial orbit rendering, celestial body information display, Chinese/English switching, and online quizzes.

Technical Architecture:

Frontend

3D Rendering: Three.js (ES Module) + OrbitControls

Styling: CSS Custom Properties (Theme System)

Interaction: Native JavaScript

Architecture Pattern: CelestialSystem Base Class + Configuration-driven

Backend

Framework: Flask

LLM: DeepSeek V4 Flash (deepseek-v4-flash)

API: <https://api.deepseek.com/chat/completions>

Persistence: JSON File Storage

Dependencies: flask, flask-cors, requests

Testing

Framework**: pytest + Playwright

Types**: API Testing (Backend Routes) + UI Testing (Browser Automation)

Phase 1: CelestialSystem Base Class + Solar System 3D Orbits

Objective:

Establish a reusable celestial system base class `CelestialSystem` that renders a complete Solar System 3D orbit scene through configuration-driven approach.

Features:

CelestialSystem Base Class

'static/js/celestial-system.js' — Universal celestial system base class, supporting:

Core Rendering

Central Body: Rotation + Configurable texture/size/color

Orbiting Bodies: Revolve around central body + rotation, configurable orbit radius, speed, size

Axial Tilt: Achieved via 'Group.rotation.x' (e.g., Earth 23.5°, Uranus 97.8°)

Orbit Lines: Drawn with EllipseCurve, configurable color

Starfield Background: BufferGeometry + Points, randomly scattered particles

Animation Control

'speedMultiplier' speed control

'animate()' loop updates revolution angle + rotation angle + hook callbacks

Phase 2: InfoPanel + Chinese/English Switching + Celestial Selector

Objective:

Implement left-side information panel (celestial data display), global Chinese/English switching mechanism, and bottom celestial selector (including moon selection).

Features:

1. LanguageManager

`static/js/language.js` — Singleton pattern, global language state management

2. InfoPanel

`static/js/info-panel.js` — Pure JS dynamic rendering, no static HTML template

Phase 3: QuizPanel Quiz System

Objective:

Implement right-side quiz panel supporting two modes: local question bank and LLM online generation. Online questions can be saved to local bank.

Features:

QuizPanel Module

``static/js/quiz-panel.js`` — Dual-mode quiz panel

Two Modes

Mode	Data Source	Description
Local Bank	<code>/api/quiz/{topic}</code>	Shuffled display, shows "All questions completed" when done
Online Generation	<code>/api/quiz/generate</code>	DeepSeek V4 Flash generation, can continue generating

Phase 4: ChatPanel Conversation Panel + Context Memory

Objective:

Implement AI conversation panel with top search box as entry point, supporting real-time conversation with DeepSeek, with server-side context memory maintenance.

Features:

ChatPanel Module

`static/js/chat-panel.js` — Search box + conversation popup panel

Interaction Flow

1. Click top search box → Popup conversation panel
2. Enter question in search box or panel input → Enter or click send
3. Show "🤔 Thinking..." → Call LLM → Show AI response
4. Click panel close button / click outside panel → Close panel

Phase 5: Mercury + Celestial Scene Switching Framework

Objective:

Based on the Solar System panorama, implement **scene switching** after clicking celestial icons — centered celestial display, close-up observation, with texture rendering. Establish a universal celestial body addition process that all subsequent planets/moons will follow.

Use **Mercury** as the first example to verify the feasibility of the entire framework.

Phase 6: Venus + Texture Generation with Outer Layer Effects

Objective:

Based on the framework established in [Phase 5](05_phase5_mercury.md), implement Venus. Verify the framework's support for ****non-standard texture requirements**** (outer cloud layer) and ****special rotation behavior**** (retrograde rotation).

Phase 7: Earth + Moon Satellite System

Objective:

ased on the framework established in [Phase 5](05_phase5_mercury.md), implement Earth. Verify the framework's support for ****satellite systems**** and ****multi-layer texture overlay**** (surface + clouds).

Differences from Previous Celestial Bodies

Feature	Mercury	Venus	Earth
Moons	None	None	Moon (1)
Clouds	None	Outer cloud layer (semi-transparent sphere)	Inner clouds (mesh sub-object)
Axial Tilt	0.034°	177.4°	23.44°
Radius	0.95	0.95	1.0
Texture	Gray craters	Orange-yellow cloud bands	Blue oceans + green continents + ice caps

Phase 8: Mars + Thin Atmosphere + Satellite System

Objective:

Based on the framework established in [Phase 5](05_phase5_mercury.md), implement Mars. Verify the framework's support for **thin atmosphere** (sub-object attachment) and **asteroid-type satellites**.

Differences from Previous Celestial Bodies

Feature	Mercury	Venus	Earth	Mars
Moons	None	None	Moon (1)	Phobos + Deimos (2)
Atmosphere	None	Outer cloud layer	Inner clouds (sub-object)	Thin atmosphere (sub-object)
Axial Tilt	0.034°	177.4°	23.44°	25.19°
Radius	0.95	0.95	1.0	0.95
Texture	Gray craters	Orange-yellow clouds	Blue oceans + green continents	Reddish-brown + polar caps + canyons + volcanoes + craters

Phase 9: Jupiter + Gas Giant + Galilean Moons System

Objective:

ased on the framework established in [Phase 5](05_phase5_mercury.md), implement Jupiter. Verify the framework's support for **gas giants** (large radius, fast rotation) and **4 Galilean moons**.

Differences from Previous Celestial Bodies

Feature	Mercury	Venus	Earth	Mars	Jupiter
Planet Type	Rocky	Rocky	Rocky	Rocky	Gas Giant
Moons	None	None	1	2	4 Galilean moons
Clouds	None	Outer cloud layer	Inner clouds	Thin atmosphere	Clouds (centerGroup)
Axial Tilt	0.034°	177.4°	23.44°	25.19°	3.13°
Radius	0.95	0.95	1.0	0.95	1.0 (max display)
Rotation Speed	Slow	Retrograde medium	Medium	Medium	Very fast (rotationSpeed 0.015)
Texture	Gray craters	Orange-yellow clouds	Blue oceans	Reddish geology	Brown/beige stripes + Great Red Spot + vortices

Phase 10: Saturn + Particle Ring System + 7 Moon System

Objective:

Based on the framework established in [Phase 5](05_phase5_mercury.md), implement Saturn. Verify the framework's support for **particle ring system** (multiple ring bands + gaps), **Z-axis tilt rotation**, and **large moon system** (7 moons).

Differences from Previous Celestial Bodies

Feature	Mars	Jupiter	Saturn
Planet Type	Rocky	Gas Giant	Gas Giant + Rings
Moon Count	2	4	7
Rings	None	None	7-layer particle ring system (C/B/A/F + Cassini Division + Encke Gap)
Tilt Axis	X-axis (default)	X-axis (default)	Z-axis
Axial Tilt	25.19°	3.13°	26.73°
Radius	0.95	1.0	2.5 (max in close-up)
Camera Distance	z: 7	z: 8	z: 35 (distant for ring view)
Texture	Reddish geology	Brown/beige stripes	Pale yellow/beige + atmospheric bands + storm spots + polar regions

Phase 11: Uranus + Sideways Rotation + 5 Moon System

Objective:

Based on the framework established in [Phase 5](05_phase5_mercury.md), implement Uranus. Verify the framework's support for **extreme axial tilt** (97.77°, sideways rotation), **atmosphere sub-object**, and **5 moon system**.

Differences from Previous Celestial Bodies

Feature	Jupiter	Saturn	Uranus
Planet Type	Gas Giant	Gas + Rings	Ice Giant
Axial Tilt	3.13°	26.73° (Z-axis)	97.77° (nearly lying on side)
Moon Count	4	7	5 (Miranda~Oberon)
Atmosphere	Clouds(centerGroup)	None	Atmosphere(centerMesh sub-object)
Rings	None	7-layer particle rings	None (not implemented this phase)
Radius	1.0	2.5	1.0
Camera Distance	z: 8	z: 35	z: 10 (view all 5 moons)
Texture	Brown/beige stripes	Pale yellow + storms	Cyan-blue + faint clouds + dark spots + polar regions

Phase 12: Neptune + Great Dark Spot + Triton Retrograde Orbit

Objective:

Based on the framework established in [Phase 5](05_phase5_mercury.md), implement Neptune (the last planet in the Solar System). Verify the framework's support for **ice giant textures** (deep blue tones + Great Dark Spot) and **single moon retrograde orbit**.

Differences from Previous Celestial Bodies

Feature	Uranus	Neptune
Planet Type	Ice Giant	Ice Giant
Axial Tilt	97.77° (sideways)	28.32° (similar to Earth)
Moon Count	5	1 (Triton)
Moon Orbit Direction	Prograde	Retrograde (clockwise)
Atmosphere	Sub-object attach	Independent object (scene.add)
Texture Features	Cyan-blue + faint clouds	Deep blue + Great Dark Spot + bright white storms
Radius	1.0	0.95
Camera Distance	z: 10	z: 9
Texture	Pale cyan-blue	Deep rich blue

Phase 13: Moon System Framework + Moon

Objective:

Launch Phase 1 of the moon texture system. Verify `CelestialSystem.addOrbitingBody()` support for moon `textureGenerator` (framework already supports, no changes needed), implement Moon texture close-up first. Establish standard process for all subsequent moons.

Framework Capability Verification

- 1.CelestialSystem Already Supports Moon Textures
- 2.Moon Texture Generation Pattern

Phase 14: Jupiter's Galilean Moons (4)

Objective:

Based on the moon framework from Phase 13, implement close-up modes for Jupiter's 4 Galilean moons. Verify framework support for diverse texture types (sulfur volcanoes, ice crust cracks, ice-rock mix, ancient craters).

Overview of 4 Galilean Moons

Moon	English Name	Visual Features	Color
Io	Io	Sulfur yellow + orange deposits + red volcanoes + white plumes + dark calderas + mottled	#ffffcc
Europa	Europa	Ice white base + dark ice cracks + parallel ridges + dark/bright regions + craters	#ccddff
Ganymede	Ganymede	Gray-brown ice-rock mix + dark ancient regions + bright groves + craters + ice cracks	#aaaaaa
Callisto	Callisto	Gray-brown ancient surface + Valhalla multi-ring structure + 400 dense craters + mottled texture + ejecta rays	#886644

Phase 15: Saturn's Moon System (7)

Objective:

Implement close-up modes for Saturn's 7 major moons with independent texture generation.

Overview of 7 Moons

Moon	English Name	Visual Features	Texture Source
Mimas	Mimas	Gray ice crust + Herschel crater (1/3 diameter) + central peak	Independent `mimas.js`
Enceladus	Enceladus	Bright white ice crust + tiger stripe cracks + high reflectivity	Independent `enceladus.js`
Tethys	Tethys	Light gray ice moon + craters	Shared `saturn-ice-moons.js` (light)
Dione	Dione	Medium gray ice moon + bright spots	Shared `saturn-ice-moons.js` (medium)
Rhea	Rhea	Dark gray ice moon + craters	Shared `saturn-ice-moons.js` (dark)
Titan	Titan	Orange-yellow thick atmosphere + methane lakes + dunes	Independent `titan.js`
Iapetus	Iapetus	Two-tone (light/dark) + craters	Shared `saturn-ice-moons.js` (twoface)

Phase 16: Uranus' 5 Moons + Triton (6)

Objective:

Implement close-up modes for Uranus' 5 major moons and Neptune's Triton. Complete all moon textures except Phobos/Deimos.

Overview of 6 Moons

Moon	English Name	Parent Planet	Visual Features	Texture Source
Miranda	Miranda	Uranus	Gray ice-rock + chevron terrain + craters	`uranus-ice-moons.js`
Ariel	Ariel	Uranus	Bright gray ice moon + canyon cracks + bright spots	`uranus-ice-moons.js`
Umbriel	Umbriel	Uranus	ery dark gray + few bright crater rims	`uranus-ice-moons.js`
Titania	Titania	Uranus	Medium gray (largest Uranus moon) + craters	`uranus-ice-moons.js`
Oberon	Oberon	Uranus	Dark gray + ancient cratered surface	`uranus-ice-moons.js`
Triton	Triton	Neptune	Pink-gray cantaloupe terrain + nitrogen ice plumes + ice caps	Independent `triton.js`

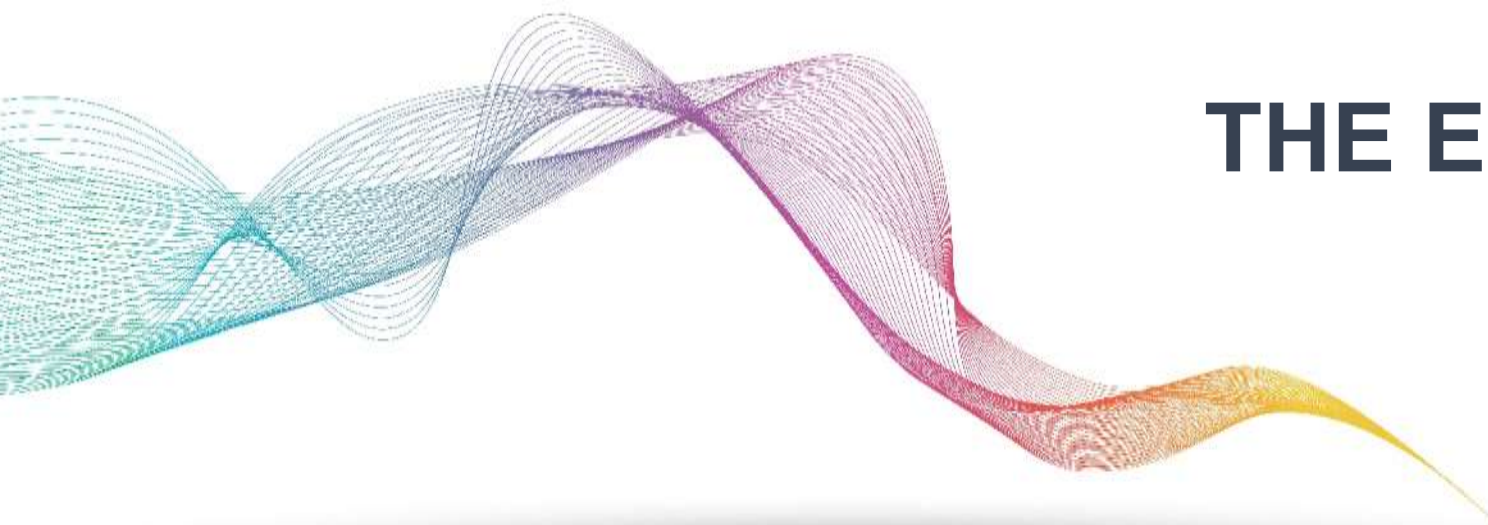
Phase 17: Phobos + Deimos — Moon Close-up Completion

Objective:

Implement close-up modes for Mars' two moons (Phobos / Deimos), completing texture coverage for all 21 moons in the Solar System.

Phobos / Deimos Overview

Moon	English Name	Visual Features	Color
Phobos	Phobos	Gray-brown irregular shape + Stickney crater (1/3 diameter) + dense grooves + rough surface	`#8b7355`
Deimos	Deimos	Gray-brown smooth surface + fewer craters + fine-grained texture	`#a0826d`



THE END